

Architecture Justification

Smart Parking Space Rental & License Validator

1. Architectural Style Analysis

Layered	Separates the interface, business logic and data. Suitable for a university Smart Parking prototype.
Microservices	Supports independent scaling and deployment, but adds network, deployment and data-consistency complexity.
Client-Server	Centralizes requests at a server. Its internal responsibilities still need a design, such as layers; the styles can coexist.

2. Architecture Selection

We chose Layered Architecture for the Smart Parking Space Rental & License Validator system. It has a Presentation Layer, a Business Logic Layer and a Data Layer within one application.

3. Reason 1 — Separation of Responsibilities

The Vehicle Owner Interface and Parking Warden Dashboard handle user interaction. Reservation management, license validation, session billing and overstay monitoring belong to the business layer. The data layer stores parking, reservation and session information. This separation makes each part easier to test, change and maintain.

4. Reason 2 — Appropriate Complexity

The parking functions share reservation and session data. A layered university prototype keeps these functions logically separate without the separate deployments and distributed data coordination required by microservices. This makes implementation and maintenance manageable.

5. Security Advantage

The interfaces and gate device access parking data through controlled business components rather than directly querying the database. Access checks can restrict vehicle and reservation details to authorized users, and a session accepts entry only after successful reservation validation. These controls reduce data exposure; layers alone do not guarantee security.

6. Performance Benefit

Plate verification follows a focused path: License Plate Validation Service → Reservation Manager → Parking Database. Keeping these components in one application avoids inter-service network calls. Indexed plate and active-reservation lookups can help meet the under-300-ms verification requirement; response time must still be measured under load.

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UML Component Diagram — Layered Architecture

